FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. AB

Sol. As shown in fig. for reflection from the ceiling

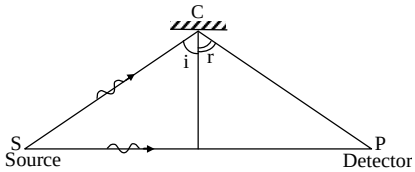
Path SCP = SC + CP = 2SC

[as $\angle i = \angle r$, SC = CP]

path SCP = $2\sqrt{60^2+25^2}$ = 130 m

So path difference between interfering waves along paths SCP and SP

$\Delta x = 130 - 120 = 10$ m



Now for constructive interference at P

$\Delta x = n\lambda$, i.e. $10 = n\lambda$

or $\lambda = \frac{10}{n}$ with n = 1,2,3.....

i.e. λ = 10m, 5m, (10/3) m and so on

2. BCD

Sol. Let,

C_V' = molar heat capacity of the first gas,

C_V'' = molar heat capacity of the second gas,

C_V = molar heat capacity of the mixture,

and similar symbols for other quantities. Then,

$\gamma = \frac{C'_P}{C'_V} = 1.67$

and

$C'_P = C_V' + R$

This gives

$C_V' = 3/2R$ and $C_P' = 5/2\ R$.

Similarly, $\gamma = 1.4$ gives $C_V'' = 5/2R$

and $C_P'' = 7/2R$

Suppose the temperature of the mixture is increased by dT. The increase in the internal energy of the first gas $= n_1 C_V' dT$.

The increase in internal energy of the second gas $= n_2 C_V'' dT$.

Thus,

$$(n_1 + n_2)C_V dT = n_1 C_V' dT + n_2 C_V'' dT$$

or, $C_V = \frac{n_1 C_V' + n_2 C_V''}{n_1 + n_2}$ ...(1)

$$C_P = C_V + R = \frac{n_1 C_V' + n_2 C_V''}{n_1 + n_2} + R$$

$$= \frac{n_1(C_V' + R) + n_2(C_V'' + R)}{n_1 + n_2}$$

$$= \frac{n_1 C_P' + n_2 C_P''}{n_1 + n_2} \quad ...(2)$$

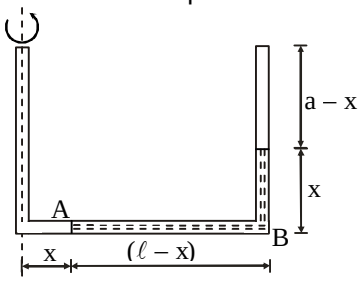
From (1) and (2),

$$\gamma = \frac{C_P}{C_V} = \frac{n_1 C_P' + n_2 C_P''}{n_1 C_V' + n_2 C_V''}$$

$$= \frac{4 \times \frac{5}{2}R + 2 \times \frac{7}{2}R}{4 \times \frac{3}{2}R + 2 \times \frac{5}{2}R} = 1.54$$

3. A

Sol. When tube is rotated, liquid starts to flow radially outward and air in sealed arm is compressed. Let the shift of liquid be x as shown in figure.



Let cross-sectional area of tube be S. Initial volume of air, $V_0 = Sa$ and initial pressure

$P_0 = 10500\ Nm^{-2}$

Final volume, $V = S(a - x)$

$\therefore$ Final pressure, $P = \frac{P_0 V_0}{V} = \frac{P_0.a.}{(a - x)}$

or Pressure at B, $P_2 = P + x\rho g = \frac{P_0 a}{(a-x)} + x\rho g$

Centripetal force required for circular motion of vertical column of height x of liquid is provided by reaction of the tube while that to horizontal length (l –x) is provided by excess pressure at B.
Force exerted by pressure difference is

$F_1 = (P_B - P_A)\, S = (P_2 - P_0)$

$S = \left[\frac{P_0 x}{(a-x)} + x\rho g\right]$

Mass of horizontal arm AB of liquid is,

$m = S(l - x)\rho$

Radius of circular path traced by its centre of mass is $r = x + \frac{\ell - x}{2} = \left(\frac{\ell + x}{2}\right)$

$\therefore$ Centripetal force, $F_2 = m\omega_0^2 r$

But $F_2 = F_1$ or

$S\rho(l-x)\left\{\frac{\ell + x}{2}\right\}\omega_0^2 = \left\{\frac{P_0 x}{(a-x)} + x\rho g\right\} S$

or x = .01 m = 1 cm

$\therefore$ Length of air column in sealed arm=(a – x)
= 5 cm Ans.

4. BC

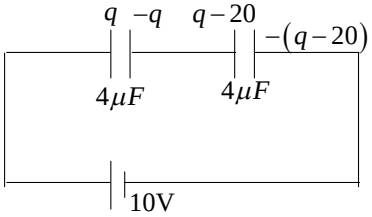
Sol. Current $I = \frac{2mV}{\left(m^2+2\right)r}$

On maximizing $m = \sqrt{2}$

$I_{max} = \frac{V}{\sqrt{2}r}$

5. BC

Sol. $\frac{q}{4} + \frac{1}{2}(q-20) = 10$



$q + 2q - 40 = 40\mu C$

$3q = 80\mu C \Rightarrow q = \frac{80}{3}\mu C$

Charge flow $= \frac{80}{3} - 20 = \frac{20}{3}\mu C$

$\omega_{battery} = \frac{20}{3} \times 10 = \frac{200}{3}\mu J$

6. AC

Sol. Magnetic moment $=IA$

Torque $=\vec{m}\times\vec{B}$

7. ACD

Sol. Electric field at distance x from the center (outside the region) is given by

$E=\frac{kR^2}{2x}\Rightarrow$ force required $F=\frac{kqR^2}{2x}$

Take line integral of dot product of force and displacement along a straight line from point P of infinity.

8. AC

Sol. $P_{glass}=-\frac{1}{12}$

$P_{water}=\frac{2}{45}$

$P_{mirror}=\frac{2}{15}$

$P_{combination}=-2P_{lens}+P_m$

$=\frac{2}{12}-\frac{4}{45}-\frac{2}{15}=\frac{-10}{180}=\frac{-1}{18}$

$f_{combination}=-18cm$

Focal length is negative hence it behaved like a concave mirror.

9. B

Sol. It box contains a 4Ω resistor equivalent resistance $=3\Omega$

Current $I=4A$ equally divided in both branches

P.d. across $AB=2\times2=4V$

10. A

Sol. In steady state current through upper branch = 0

$I=\frac{12}{6}=2A$

$P.d.$ across $AB=4V$

11. C

Sol. Potential at every point on sphere = 0

12. A

Sol. Potential due to surface charges at any point on the sphere = - potential due to point charge q at that point.

SECTION – C

13. 00002.00

Sol. Using conservation of mechanical energy $\frac{1}{2}m\left(10\sqrt{2}\right)^2=\frac{kq_1q_2}{2\times(25cm)}$

$$\frac{1}{2}mV^2{}_{\max} = \frac{kq_1q_2}{25cm}$$

$$\therefore V_{\max}^2 = 2\times\left(10\sqrt{2}\right)^2$$

$$V_{\max} = 20m/s$$

14. 00002.00

Sol. $\tau_1 = RC$

$$\tau_2 = \frac{L}{R}$$

$$\Rightarrow LC = t_1t_2 = 0.1\text{sec}$$

$$\Rightarrow T = 2\pi\sqrt{LC} = 2\pi\sqrt{\frac{1}{10}} = 2\,\text{sec}$$

15. 00002.00

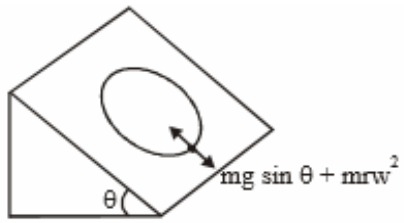
Sol. $$Heat = 700\times10^3\times1.6\times10^{-19}\times\frac{dn}{dt} = 10\times10^{-3}$$

$$\frac{dN}{dt} = \frac{10^{-1}}{10^{-14}}\times\frac{1}{7\times16} = \frac{10^{12}}{11.2} = \lambda N_0$$

$$\lambda = \frac{\ln 2}{14\times86400} \Rightarrow N_0 = \frac{14\times86400\times10^{12}}{11.2\ln 2} = 154\times10^{15}$$

16. 00006.00

Sol. $f \geq mg\sin\theta + mrw^2$



$$\mu\, mg\cos\theta \geq mg\sin\theta + mrw^2$$

$$\text{or } \mu \geq \tan\theta + \frac{w^2r}{g\cos\theta}$$

17. 00004.00

Sol. $$d\phi = \frac{\mu_0 i}{2\pi x}\times l\,dx$$

$$\phi = \frac{\mu_0 i}{2\pi} l\,\ln\left(\frac{d_2}{d_1}\right)$$

$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0}{2\pi} l\,\ln\left(\frac{d_2}{d_1}\right)\times i_0\omega\,\cos wt$$

$$\varepsilon_{rms} = \frac{\mu_0}{2\pi} l \ln\left(\frac{d_2}{d_1}\right)\frac{i_0\omega}{\sqrt{2}}$$

$$= \frac{4\pi\times10^{-7}}{2\pi}\times 1\times \ln 2\times\frac{2\sqrt{2}}{\ln 2}\times\frac{50}{\sqrt{2}}$$

18. 00004.00

Sol. The block slips when the static friction force reaches its maximum value $f_{s\,max} = \mu_s n = \mu_s mg$ .This occurs when the train has moved

$$\Delta x = \frac{f_{smax}}{k} = \frac{\mu_s mg}{k} = \frac{(0.60)(2.0\text{kg})(10\text{m/s}^2)}{50\text{N/m}} = 24\,\text{cm}$$

The time at which the block slips is

$$t = \frac{\Delta x}{v} = \frac{24\text{cm}}{6\,\text{cm/s}} = 4\text{s}$$

Chemistry

PART – II

SECTION – A

19. BC

20. ABCD
Sol. Froth floatation method is used for the sulphide ores.

21. ABC
Sol. $LiCl$ has more covalent character than $NaCl$

22. ACD
Sol. A metal that crystallizes in BCC structure has a coordination number of 8.

23. ABC
Sol. For alkaline solution PH>POH

24. AC
Sol. Fact

25. ABC
Sol. $\mathrm{Ag}^{+}_{(\mathrm{aq})} + e^{-} \rightarrow \mathrm{Ag(s)}$ and $\mathrm{AgCN(s)}$

26. ABD

27. A

28. B

29. C
Sol. Inert gas added at constant volume, equilibrium will not be affected.

30. C
Sol. $Kp = K_c \left(RT\right)^{\Delta ng}$

SECTION – C

31. 00252.00
Sol. $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} \times \frac{P_1}{P_2}$

Or $\frac{n_1}{t_1} \times \frac{t_2}{n_2} = \sqrt{\frac{M_2}{M_1}} \times \frac{P_1}{P_2}$

Or $\frac{1}{38} \times \frac{57}{1} = \sqrt{\frac{M}{28}} \times \frac{0.8}{1.6}$

$\therefore \quad M = 252$

32. 00167.20
Sol. $\frac{1}{2} H_2\left(g\right) + \left(Aq\right) \rightarrow H^{+}\left(aq\right); \qquad \Delta H^{\circ} = 0 \ldots\ldots\ldots\ldots\ldots\ldots (\mathrm{i})$

$\frac{1}{2}H_2(g)+\frac{1}{2}Cl_2(g)\rightarrow HCl(g);\qquad \Delta H^\circ=-92.4kJ.....(ii)$

$HCl(g)+H_2O(l)\rightarrow H_3O^+(aq)+Cl^-(aq);\qquad \Delta H=-74.8kJ.....(iii)$

By inspection method: $\text{Eqs.}(ii)+(iii)-(i)$ reveals that

$\frac{1}{2}Cl_2(g)+(Aq)\rightarrow Cl^-(aq);\qquad \Delta H=-167.2kJ$

i.e. Heat of formation of $Cl^-(aq)=-167.2kJ$

33. 00019.17

Sol. Let solubility of $PbBr_2$ is $S\ mol\ litre^{-1}$

$$PbBr_2(s)+Aq\rightleftharpoons PbBr_2(aq)\rightleftharpoons \underset{\frac{S\times80}{100}}{Pb^2}+\underset{\frac{2S\times80}{100}}{2Br^-}$$

Since, $PbBr_2$ ionises to 80% only.

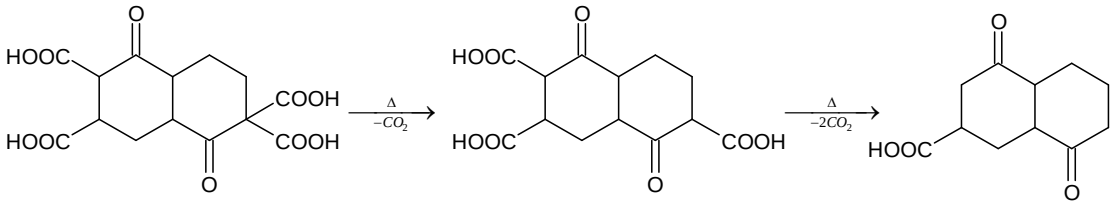
Now $\qquad K_{sp}=\left[Pb^{2+}\right]\left[Br^-\right]^2$

Or $\qquad 3.2\times10^{-5}=\left[\frac{S\times80}{100}\right]\left[\frac{2S\times80}{100}\right]^2$

$$S=0.025\,mol\,litre^{-1}$$
$$=0.025\times367\,g\,litre^{-1}$$
$$=19.175\,g\,litre^{-1}$$

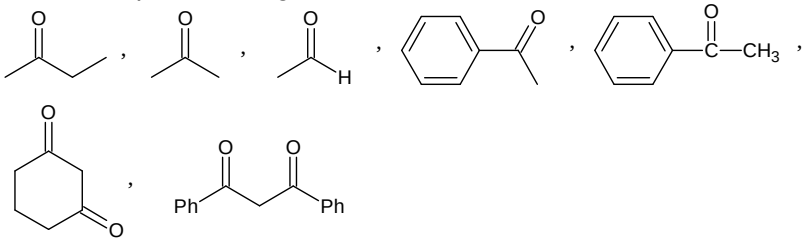
34. 00003.00

Sol.



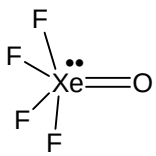
35. 00006.00

Sol. These compounds will give iodoform test



36. 00004.00

Sol. $XeOF_4$



Mathematics

PART – III

SECTION – A

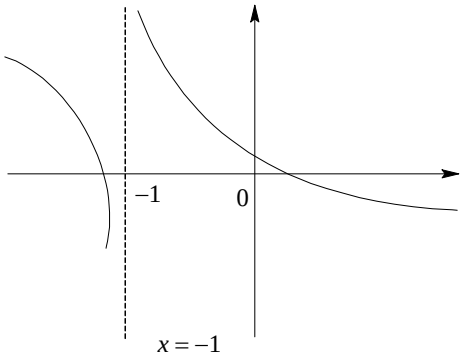
37. AD

Sol. Required probability $= \dfrac{{}^{n}C_1 \times \dfrac{n-1}{2}C_2}{{}^{n}C_3}\left(n \text{ is even}\right)$

And if n is odd $\dfrac{{}^{n}C_1 \times \dfrac{n-1}{2}C_2}{{}^{n}C_3}$

38. ABC

Sol. $f'(x) = \dfrac{-5}{(x+1)^6} - 7 + 2\cos x < 0 \ \forall x \in R$



39. AB

Sol. $P = \dfrac{1}{2}\cos x\left(\sin\left(y+z\right)+\sin\left(y-z\right)\right) \geq \dfrac{1}{2}\cos x \sin\left(y+z\right) = \dfrac{1}{2}\cos^2 x$

But, $x = \dfrac{\pi}{2} - \left(y+z\right) \leq \dfrac{\pi}{2} - 2.\dfrac{\pi}{12} = \dfrac{\pi}{3}$

$\therefore P \geq \dfrac{1}{8}$

$P = \dfrac{1}{2}\cos z\left(\sin\left(x+y\right)-\sin\left(x-y\right)\right)$

Again $P \leq \dfrac{1}{2}\cos^2 z = \dfrac{1+\cos 2z}{4}$

$P \leq \dfrac{2+\sqrt{3}}{8}$

40. BC

Sol. z_1, z_2 be the two points on the given line then

$\dfrac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} = -\dfrac{a}{a} \qquad \ldots(1)$

Also, $\frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} = \pm i \frac{z - c}{\bar{z} - \bar{c}}$ ....(2)

From (1) and (2), we get $\frac{z-c}{a} \pm i \frac{z-c}{a} = 0$

41. AB

Sol. $|\lambda - 5| = 4$

42. BCD

Sol. The line y = mx + c, passes through centroid of $\Delta C_1C_2C_4$ and $\Delta C_2C_3C_5$

$\Rightarrow y = x - 1 \Rightarrow m = 1, c = -1$

43. D

Sol. We have, $\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix}$

$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \times \begin{vmatrix} -a & b & c \\ -b & c & a \\ -c & a & b \end{vmatrix}$

$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2$

$= \left[a\left(bc - a^2\right) + b\left(ac - b^2\right) + c\left(ab - c^2\right) \right]^2$

$= \left[a^3 + b^3 + c^3 - 3abc \right]^2$

$k = -3$

44. CD

Sol. When no restriction on i and j

$S = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{1}{a^i a^j} = \left(1 + \frac{1}{a} + \frac{1}{a^2} + ...\infty \right)^2 = \frac{a^2}{(a-1)^2}$

$\Rightarrow S = \frac{\lambda a^2}{(a-1)^2 (a+1)} \Rightarrow \lambda > 2$

45. C

46. B

Sol. **(for Q. No. 45 – 46)**

(I) $e_1 < 1 < e_2$ P $f(1) < 0 P\ a > 3$

(II) Both roots are greater than $1P\ D^3\ 0P|a|^3\ 2$ and $f(1) > 0P\ a < 3$ and also

$\frac{a}{2} > 1P\ a > 2P\ a\hat{I}(2,3)$

(III) $\frac{1}{e_1^2}+\frac{1}{e_2^2}=1\text{P}\ \frac{a^2-4}{4}=1\text{P}\ a=\pm 2\sqrt{2}\ \&\ \text{D}>0\text{P}\left|a\right|\geq 2$

(IV) $e_2<\sqrt{2}<e_1 P\ f\left(\sqrt{2}\right)<0\,\text{P}\ a>2\sqrt{2}$

Equation of such ellipse is $x^2+4y^2-4+l\left(x-1\right)=0$ with area $=\frac{3P}{2}$

47. B

48. A

SECTION – C

49. 00000.32

50. 00031.00

Sol. $\frac{\left(\text{x}-3\right)^{\frac{-|\text{x}|}{\text{x}}}\sqrt{\left(\text{x}-4\right)^2}\ \left(17-\text{x}\right)}{\sqrt{-\text{x}}\left(-\text{x}^2+\text{x}-1\right)\left(|\text{x}|-32\right)}<0$

For $\sqrt{-x}$ to be defined $x<0$

$\Rightarrow \frac{-|x|}{x}=1$

Since $\sqrt{\left(x-4\right)^2},\sqrt{-x}$ are positive & $-x^2+x-1$ is always negative

$\frac{\left(x-3\right)\left(x-17\right)}{x+32}>0$

$x<0$

$\Rightarrow \frac{1}{x+32}>0\Rightarrow x=31$

51. 00833.33

Sol. $\because\ \int_a^x e^{f(t)}dt=\int_0^x g(x-t)dt+2x+3$

$\Rightarrow\ \int_a^x e^{f(t)}dt=\int_0^x g(t)dt+2x+3$ (Using King Property)

Differential both sides, we get

$e^{f(x)}=g(x)+2$

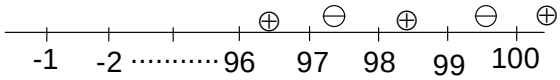
$\Rightarrow g(x)=e^{f(x)}-2$

$\Rightarrow g'(x)=e^{f(x)}.f'(x)$

$\because e^{f(x)}$ is always greater than zero.

$\therefore$ Sign of g'(x) is same as sign of $f'(x)$.

$\therefore$ Sign of g'(x)



Clearly, local extremum (maximum or minimum) will occur at
x=99,97,95,….,3,1

$\therefore$ Sum of all the values=1+3+5+…+99= $\frac{50}{2}[2\times1+(50-1)\times2]=2500$

52. 00008.00

Sol. The triangle has circumcentre at origin and its orthocentre laying on the circumcircle.

53. 00005.00

Sol. $\lim\limits_{y\to1}\frac{100y^{100}}{y^{100}-1}-\frac{50.y^{50}}{y^{50}-1}$

$\lim\limits_{y\to1}100-50-\left(\frac{100}{1-y^{100}}-\frac{50}{1-y^{50}}\right)$

L=50-L $\therefore$ L= $\frac{50}{2}=25$

54. 00001.00

Sol. $T_n=\frac{H_{n+1}-H_n}{H_nH_{n+1}}$

As $H_{n+1}-H_n=\frac{1}{n+1}$

$\Rightarrow T_n=\frac{1}{H_n}-\frac{1}{H_{n+1}}$

$\Rightarrow T_1=\frac{1}{H_1}-\frac{1}{H_2}$

$T_2=\frac{1}{H_2}-\frac{1}{H_3}$

⋮ ⋮ ⋮

$T_n=\frac{1}{H_n}-\frac{1}{H_{n+1}}$

$T_1+T_2+...+T_n=\frac{1}{H_1}-\frac{1}{H_{n+1}}$

As $n\to\infty$, $T_1+T_2+...+0=1$